One-Way Between Groups or Independent ANOVA

When to Use

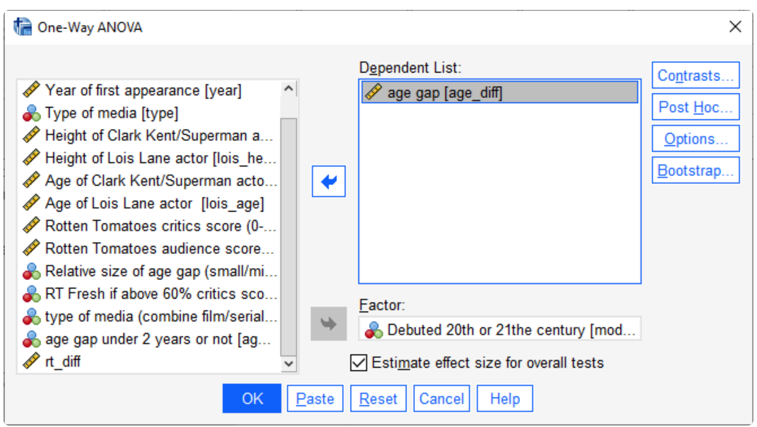
To compare means of a quantitative variable obtained from 2 independent groups.

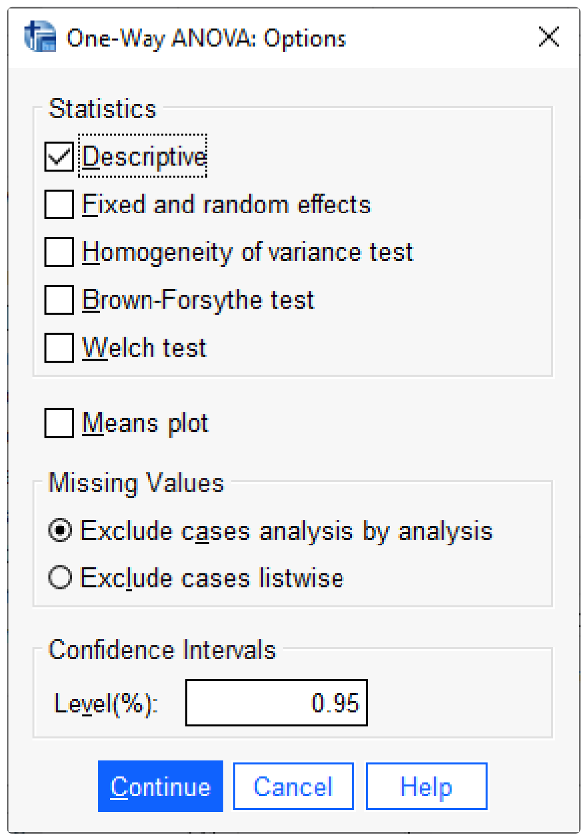
Research Hypothesis: Superman media from the 21st century will have a larger average age gap between actors playing Lois Lane and Clark Kent than media from the 20th century.

Null Hypothesis: There is no mean difference in age gap between Superman media from the 20th and 21st centuries.

Analyze → Compare Means → One-way ANOVA

- highlight the "Dependent" variable (be sure it is **quantitative**) and click the arrow
- highlight the "Factor" (IV, grouping) variable (be sure it is **qualitative**) and click the arrow
- "Options" — check that you want "Descriptive Statistics"





SPSS Syntax

```
ONEWAY age_diff BY modern            (1)
  /STATISTICSDESCRIPTIVES            (2)
  /MISSINGLISTWISE.                  (3)
```

1. Dependent variable(s) BY Factor
2. get descriptive statistics for each factor group
3. listwise deletion (alternative is "ANALYSIS")

SPSS Output

Descriptives

age_diff

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean Lower Bound	Upper Bound	Minimum	Maximum
1	4	7.19	5.30	2.65	−1.24	15.61	1.64	13.03
2	5	4.55	3.06	1.37	0.75	8.35	1.15	8.71
Total	9	5.72	4.14	1.38	2.54	8.90	1.15	13.03

Assuming the null hypothesis is true, the p-value is the probability of obtaining an F statistic at least as large as the one observed.

SPSS may display **.000** when a very small p-value rounds to three decimal places. Report that value as **p < .001**, not p = .000.

ANOVA

age_diff

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	15.453	1	15.453	0.890	.377
Within Groups	121.539	7	17.363		
Total	136.992	8			

The **F-statistic** is the ratio of between-group variance to within-group variance. A larger F indicates greater differences between group means relative to variability within groups.

The Mean Square for Within Groups is also called the Mean Square Error (MSE). You should report MSE along with F, df, and p in your results.

The F-statistic of 0.89 with df = (1, 7) indicates that there is not a significant difference between the groups.

Reporting the Results

¶ Sample APA Write-Up

The age gap for each century of debut is summarized in the table above. 20th Century (*M* = 7.19, *SD* = 5.30) and 21st Century (*M* = 4.55, *SD* = 3.06). There was no significant difference in age gap between the two groups, $F(1,7) = 0.89$, $MSe = 17.36$, $p = .377$. Contrary to the research hypothesis, there was no significant difference in age gap between 20th Century and 21st Century portrayals.

It is important to report the univariate statistics for the dependent variable for both groups before presenting the ANOVA results. Often these are presented in a table or a figure.

As in the example, be sure to communicate:

- The research hypothesis (if there is one)
- Descriptive statistics (M, SD) for each group
- The ANOVA results (*F*, *df*, *p*, *MSe*)
- The direction of the difference (which group scored higher)
- Whether or not those results support the research hypothesis

Sometimes, the univariate statistics are presented in text, along with the ANOVA results.